Lab Manual

Department of Computer Science and Engineering

Faculty Name : **Ms. Kritika / Mr. Manish Kumar Jha/Dr. Sangeeta Soni**

Subject Name : **Computer Graphics and Multimedia Techniques**

Subject Code : **5CS4 - 21**

Session : 2024-2025

GLOBAL INSTITUTE OF TECHNOLOGY

(Rajasthan Technical University, KOTA

Session 2024-25

FACULTY OF GLOBAL INSTITUTE OF TECHNOLOGY.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

NAME OF FACULTY: Ms.Kritika / Mr.Manish Kumar Jha/Dr.Sangeeta Soni

NAMEOF COURSE : COMPUTER GRAPHICS AND MULTIMEDIA LAB

COURSECODE : 5CS4 - 21

LAB COURSEFILECONTENT

Course Planning (Outcome Based Education System)
1. Vision & Mission of the Institute &Department
2. Program Specific Outcomes (PSO) and PEO Program Education Objectives
3. Program Outcomes (POs)
4. RTU Syllabus
5. List of Experiment
6. Lab Course Outcomes
7. Course Articulation Matrix (CO-PO-PSO Mapping)
8. Personal Time Table
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11. Experiment and output list
12. Viva Question
13. Lab Quiz

Vision, Mission Statements of the Institute

GLOBAL INSTITUTE OF TECHNOLOGY

Vision, Mission of The Institute

Vision

We would Contribute to Human Development through Academic Pursuits and be a Trendsetter in the field of Technical Education.

Mission

To establish world-class high quality learning environment by way of developing value-based education system, powered by brilliant professionals and leaders in the field of Engineering.

Principal

Principal
Global Institute of Technology
Jaipur 302022

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Vision, Mission Statements of the Department

GLOBAL INSTITUTE OF TECHNOLOGY

Vision, Mission of the Department (Computer Science & Engineering)

Vision

The vision of the Department of Computer Science & Engineering is to be recognized as a pioneer in core competencies, quality education, multidisciplinary approaches and research.

Mission

M1: To impart fundamental knowledge of Computer Science & Engineering through quality education.

M2: To build a strong foundation in engineering projects and research through value-based education.

M3: To develop graduates who can provide leadership and service in the field of Computer Science & Engineering and Contribute to society.

Head of Department
(Computer Science & Engineering)

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Program Specific Outcome (PSO)

S. No.	PSO	Description
PSO 1	Knowledge Enhancement in Computing	Graduates demonstrate a clear understanding of computer system hardware and software and effectively apply this knowledge to analyze and solve real-world problems in the efficient design of computer systems.
PSO 2	Software Design and Development	Graduates understand the software development process and are able to design and develop software using strong practical programming skills.

Program Education Objectives (PEO)

S. No.	Description
PEO1	To develop core competencies that prepare Computer Science and Engineering graduates for successful careers as innovators, leaders, professionals, and researchers.
PEO2	Graduates of the Computer Science and Engineering program will demonstrate professional and ethical conduct, effective communication skills, a multidisciplinary approach, and the ability to relate engineering issues to broader societal perspectives.
PEO3	Graduates of the Computer Science and Engineering program will recognize the need for, and have the ability to engage in, independent and lifelong learning in the context of technological change.

Program Outcomes (POs)

PO1	**Engineering knowledge**	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	**Problem analysis**	Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/development of solutions**	Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	**Conduct investigations of complex problems**	Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	**Engineering Tool Usage**	Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	**The Engineer and The World**	Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	**Ethics**	Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	**Individual and Collaborative Team work**	Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	**Communication**	Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
PO10	**Project Management and Finance**	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	**Life-Long Learning**	Recognize the need for, and have the preparation and ability for i.) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

RTU Syllabus

RAJASTHAN TECHNICAL UNIVERSITY, KOTA

Syllabus

III Year-V Semester: B.Tech. Computer Science and Engineering

5CS4-21: Computer Graphics & Multimedia Lab

Credit: 1 **Max. Marks:100 (IA:60, ETE:40)**

0L+0T+2P **End Term Exam: 2 Hours**

SN	List of Experiments
1	Implementation of Line, Circle and ellipse attributes
2	To plot a point (pixel) on the screen
3	To draw a straight line using DDA Algorithm
4	Implementation of mid-point circle generating Algorithm
5	Implementation of ellipse generating Algorithm
6	Two Dimensional transformations - Translation, Rotation, Scaling, Reflection, Shear
7	Composite 2D Transformations
8	Cohen Sutherland 2D line clipping and Windowing
9	Sutherland – Hodgeman Polygon clipping Algorithm
10	Three dimensional transformations - Translation, Rotation, Scaling
11	Composite 3D transformations
12	Drawing three dimensional objects and Scenes
13	Generating Fractal images

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List of Experiments

S. No.	List of Experiments as per RTU Syllabus	Co & Blooms Level
1.	**To produce a single pixel and pre specified pattern on screen.**	**CO1&L6**
2.	**To implement features like changing background color, fore ground color, resizing window, repositioning of window**	**CO1& L3**
3.	**Write a program to implement Bresenham Line algorithm.**	**CO1&L3**
4.	**Write a program to implement DDA Line algorithm.**	**CO2&L3**
5.	**Write a program to implement midpoint circle algorithm.**	**CO2&L3**
6.	**Write a program to fill an ellipse.**	**CO3&L2**
7.	**Write a program to fill a polygon.**	**CO3&L2**
8.	**Write a program to hoist a flag (** 2D line clipping and Windowing).	**CO4&L4**
9	**Write a program to create balloons (**Sutherland – Hodgeman Polygon clipping Algorithm).	**CO4&L5**
10	**Write a program to implement a walking robot (**Three dimensional transformations - Translation, Rotation, Scaling).	**CO5&L5**

Course Outcomes (COs)

Course Outcomes		
Semester No:	B. Tech (V Semester)	
Course Title:	Computer Graphics & Multimedia Lab	**Course Code:** 5CS4-21
Course Outcome No.	**Course Outcomes Statement** **Students will be able to**	
5CS4-21.1	**Recall And Identify** basic graphics algorithms such as line, circle, and ellipse drawing.	
5CS4-21.2	**Explain** 2D geometric transformations including translation, scaling, rotation, and reflection.	
5CS4-21.3	**Apply** polygon filling techniques such as flood-fill, boundary-fill, and scan-line algorithms to compare their performance and usage.	
5CS4-21.4	**Examine** clipping algorithms like Cohen-Sutherland and Sutherland-Hodgman to **Analyze** their application for lines and polygons.	
5CS4-21.5	Analyze animation and multimedia applications and **judge** the efficiency of applied compression techniques.	

Course Articulation Matrix (CO-PO-PSO Mapping)

Course Name: Computer Graphics and Multimedia Lab												**Course Code: 5CS4 - 21**	
Course Articulation Matrix													
Course Outcomes (COs) code & Statement	**Program Outcomes (POs)**											**Program Specific Outcomes (PSOs)**	
	PO - 1	PO -2	PO -3	PO - 4	PO - 5	PO -6	PO -7	PO -8	PO -9	PO -10	PO -11	PSO- 1	PSO- 2
5CS4-21.1	3	1	1	-	2	-	-	-	-	-	1	3	1
5CS4-21.2	3	2	1	-	2	-	-	-	-	-	1	3	1
5CS4-21.3	3	2	2	1	3	-	-	-	-	-	2	3	2
5CS4-21.4	3	3	2	2	2	-	-	-	-	-	2	3	2
5CS4-21.5	3	3	3	3	1	-	-	-	-	-	2	3	2

Personal Time Table

Semester V, Section-A
MS. KRITIKA

Day\ Time	I 9:00-9:55	II 10:00-10:55	III 11:00-11:55	IV 12:00 -12:55	LUNCH 01.00-1.40	V 1:45 – 2:40	VI 2:45 – 3:40
Monday	A2 CGM Lab				LUNCH		
Tuesday							
Wednesday						A1 CGM Lab	
Thursday							
Friday							

Semester V, Section-B

Mr. Manish Kumar Jha

Day\ Time	I 9:00 - 9:55	II 10:00-10:55	III 11:00-11:55	IV 1:00 -1:40	LUNCH 01.00-1.40	V 1:45 – 2:40	VI 2:45 – 3:40
Monday					LUNCH		
Tuesday			**B2 CGM Lab**				
Wednesday							
Thursday							
Friday						B1 CGM Lab	

Semester V, Section-C
DR. SANGEETA SONI

Day\ Time	I 9:00-9:55	II 10:00-10:55	III 11:00-11:55	IV 12.00-12.55	LUNCH 01.00-1.40	V 1:45 – 2:40	VI 2:45 – 3:40
Monday					LUNCH		
Tuesday	C2 CGM Lab						
Wednesday							
Thrusday							
Friday	C1 CGM Lab						

Laboratory Time Table

Day\ Time	I 9:00-9:55	II 10:00-10:55	III 11:00-11:55	IV 12.00-12.55	LUNCH 01.00-1.40	V 1:45 – 2:40	VI 2:45 – 3:40
Monday	A2 CGM Lab				LUNCH		
Tuesday	C2 CGM Lab		B2 CGM Lab				
Wednesday						A1 CGM Lab	
Thrusday							
Friday	C1 CGM Lab					B1 CGM Lab	

General Safety Measures/Rules of Laboratory

GLOBAL INSTITUTE OF TECHNOLOGY

COMPUTER LABORATORY RULES & REGULATION

- Students are required to carry a valid ID card before entering the computer lab.
- Students are advised to maintain a quiet and respectful environment to ensure concentration for students and avoid disturbing others. Cell phones should be set to vibration or silent mode.
- Students are not allowed to bring bags, food and beverages inside the laboratory. Eating and drinking are typically prohibited in the computer laboratory.
- Students are advised to take care not to damage the computer laboratory infrastructure and/or computer or its parts or electrical fixtures. Such instances will be dealt with according to the disciplinary rules.
- Students are not allowed to alter or delete configuration settings of any computer or laboratory equipment. Tampering, deleting or modifying CMOS/BIOS settings, IP Configuration, system parameters, or system files stored in the hard disk are strictly prohibited.
- Refrain from the unauthorized hardware and peripherals modifications. Students are not allowed to modify or tamper either intentionally or unintentionally. Similarly, unplugging the lab computer power cord and Ethernet cables to connect personal laptops is prohibited.
- Students may refrain from the unethical and morally destructive usage of internet. Accessing inappropriate websites and e-resources is strictly prohibited. Use of Internet services for unwarranted purposes is strictly prohibited.
- Mobile phone use is not permitted in the labs. Students are advised to switch off their mobile phones or set them to vibration mode while using the computer lab. Students put your mobile phones in silent mode before entering the lab.
- Students are advised to turn off their computer and arrange the computer peripherals (mouse, keyboard and monitor) and chair before leaving the lab.
- The computer laboratories are under CCTV surveillance and therefore mindful of any unethical behavior. Actions will be taken on those found violating these rules and acts of indiscipline.

Head of CSE Department

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DO'S AND DONT'S

DO's

1. Enter the lab on time and leave at proper time.
2. Wait for the previous class to leave before the next class enters.
3. Keep the bag outside in the respective racks.
4. Utilize lab hours in the corresponding.
5. Turn off the machine before leaving the lab unless a member of lab staff has specifically told you not to do so.
6. Leave the labs at least as nice as you found them.
7. If you notice a problem with a piece of equipment (e.g. a computer doesn't respond) or the room in general (e.g. cooling, heating, lighting) please report it to lab staff immediately. Do not attempt to fix the problem yourself.

DONT'S

1. Don't abuse the equipment.
2. Do not adjust the heat or air conditioners. If you feel the temperature is not properly set, inform lab staff; we will attempt
3. to maintain a balance that is healthy for people and machines.
4. Do not attempt to reboot a computer. Report problems to lab staff.
5. Do not remove or modify any software or file without permission.
6. Do not remove printers and machines from the network without being explicitly told to do so by lab staff.
7. Don't monopolize equipment. If you're going to be away from your machine for more than 10 or 15 minutes, log out
8. before leaving. This is both for the security of your account, and to ensure that others are able to use the lab resources while you are not.
9. Don't use internet, internet chat of any kind in your regular lab schedule.
10. Do not download or upload of MP3, JPG or MPEG files.
11. No games are allowed in the lab sessions.
12. No hardware including USB drives can be connected or disconnected in the labs without prior permission of the lab in-
13. charge.
14. No food or drink is allowed in the lab or near any of the equipment. Aside from the fact that it leaves a mess and attracts
15. pests, spilling anything on a keyboard or other piece of computer equipment could cause permanent, irreparable, and costly damage. (and in fact, *has*) If you need to eat or drink, take a break and do so in the canteen.
16. Don't bring any external material in the lab, except your lab record, copy and books.
17. Don't bring the mobile phones in the lab. If necessary, then keep them in silence mode.
18. Please be considerate of those around you, especially in terms of noise level. While labs are a natural place for conversations of all types, kindly keep the volume turned down.

Experiment No. – 1

Aim:-To produce a single pixel and pre specified pattern on screen:

Source Code:-

```
#include<graphics.h>
#include<conio.h>

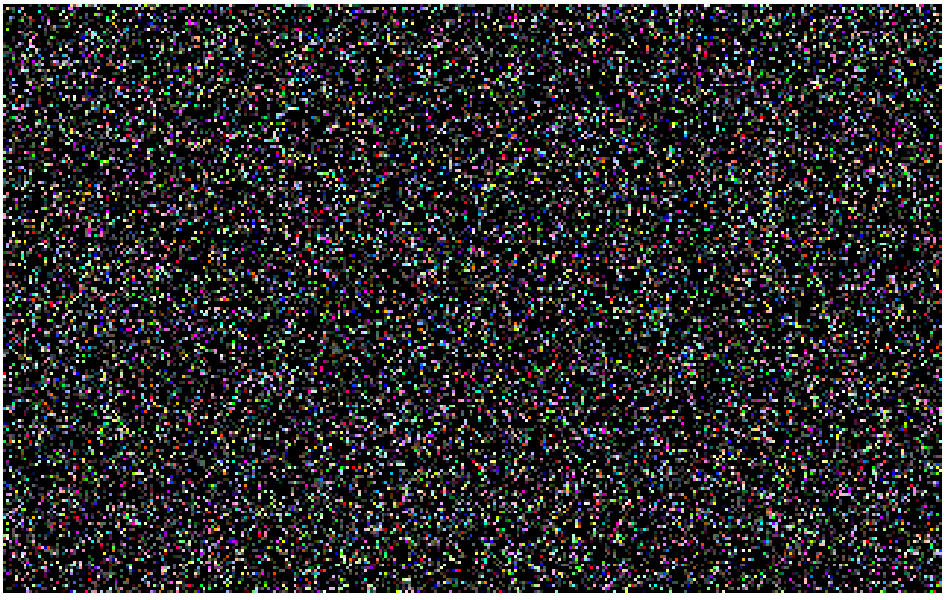
main()
{
  int gd = DETECT, gm;

  initgraph(&gd, &gm, "C:\\TC\\BGI");

  putpixel(25, 25, RED);

  getch();
  closegraph();
  return 0;
}
```

Output:-



Viva Question (Experiment No. – 1)

1. What is Pixel?
2. What is Bitmap?
3. What is Refresh Buffer/Frame Buffer?
4. How many Bits used in Bitmap?
5. What is Pixmap?

****

Experiment No - 2

Aim:- To implement features like changing background color, fore ground color, resizing window, repositioning of window

Source Code:-

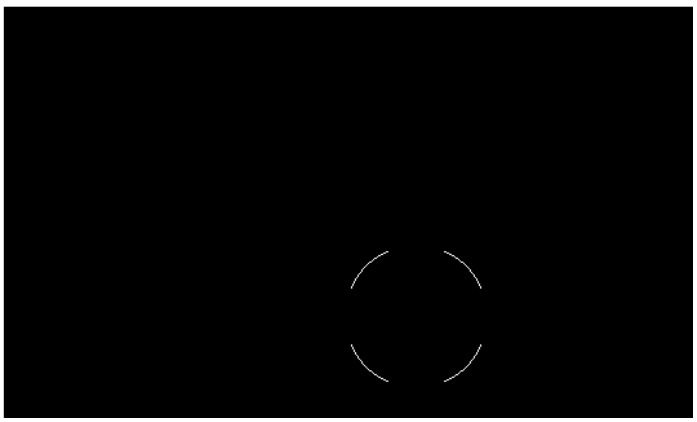
```
main()
{
  int gd = DETECT, gm;
  initgraph(&gd,&gm,"C:\\TC\\BGI");
  midx = getmaxx()/2;
  midy = getmaxy()/2;
 setbkcolor(GREEN);



 setviewport(midx - 50, midy - 50, midx + 50, midy + 50, 1);
 circle(100,100,50);          /* drawn in white color */
  setcolor(RED);
  circle(200,200,50);          /* drawn in red color   */

  getch();
  closegraph();
  return 0;
}
```

Output:-



Viva Question (Experiment No. – 2)

1. **What is Clipping?**
2. **What is the difference between Raster scan and Random scan?**
3. **How many bits are required for a 512 × 512 raster with each pixel being represented by 3 bits?**
4. **Define point clipping.**
5. **In a raster system with resolution 2560 × 2048. How many pixels could be accessed per second by a display controller that refresh the screen at a rate of 60 frames per second. Also calculate access time per pixel in the system.**

****

Experiment No - 3

Aim:-Write a program to implement Bresenham Line algorithm.

Source Code:-

```
#include<stdio.h>
#include<conio.h>
#include<graphics.h>
void main()
{
    int gd = DETECT, gm;
    int dx, dy, p, end;
    float x1, x2, y1, y2, x, y;
    initgraph(&gd, &gm, "c:\tc\bgi");
    printf("Enter Value of X1: ");
    scanf("%f", &x1);
    printf("Enter Value of Y1: ");
    scanf("%f", &y1);
    printf("Enter Value of X2: ");
    scanf("%f", &x2);
    printf("Enter Value of Y2: ");
    scanf("%f", &y2);
    dx = abs(x1 - x2);
    dy = abs(y1 - y2);
    p = 2 * dy - dx;
```

```
if(x1 > x2)
{
    x = x2;
    y = y2;
    end = x1;
}
else
{
    x = x1;
    y = y1;
    end = x2;
}
putpixel(x, y, 10);
while(x < end)
{
    x = x + 1;
    if(p < 0)
    {
        p = p + 2 * dy;
    }
    else
    {
        y = y + 1;
        p = p + 2 * (dy - dx);
    }
```

```
        putpixel(x, y, 10);
    }
    getch();
    closegraph();
}
```

Output:-

Above program draws a line from (100, 100) to (200, 200).

line program.

Output of program:



Sample Viva Question (Experiment No.3)

1. **What are the Advantages of Bresenhem's Algorithm?**
2. **What do you call the path the electron beam takes when returning to the left side of the CRT screen?**
3. **What is the refresh rate of Flicker?**
4. **Write short notes on active and passive transformations?**
5. **What do you call the path the electron beam takes at the end of each refresh cycle?**

Experiment No - 4

Aim:- Write a program to implement DDA Line algorithm.

Source Code:-

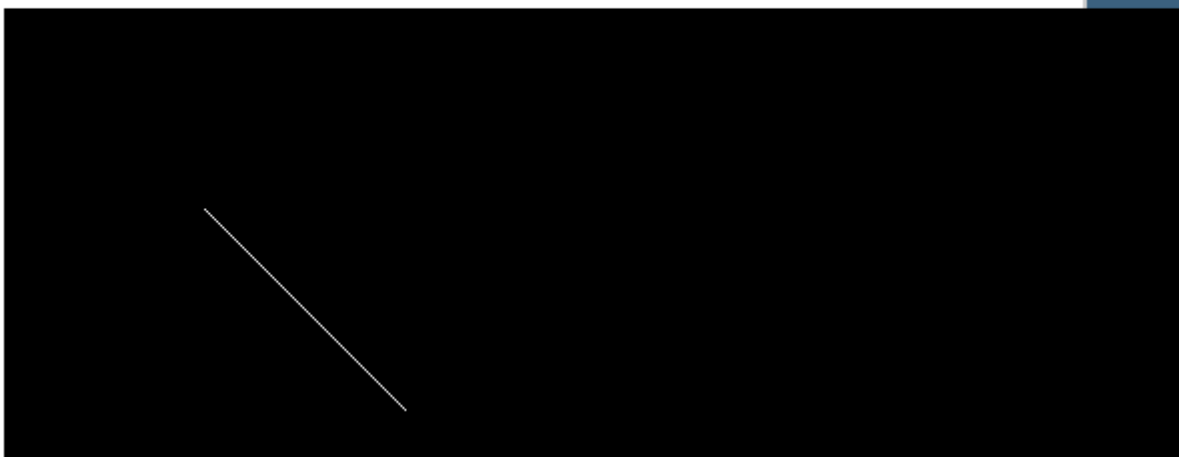
```
#include<stdio.h>
#include<conio.h>
#include<graphics.h>
#include<ctype.h>
#include<math.h>
#include<stdlib.h>
void draw(int x1,int y1,int x2,int y2);
void main()
{
int x1,y1,x2,y2;
int gdriver=DETECT,gmode,gerror;
initgraph(&gdriver,&gmode,"c:\\tc\\bgi:");
printf("\n Enter the x and y value for starting point:\n");
scanf("%d%d",&x1,&y1);
printf("\n Enter the x and y value for ending point:\n");
scanf("%d%d",&x2,&y2);
printf("\n The Line is shown below: \n");
draw(x1,y1,x2,y2);
getch();
}
void draw(int x1,int y1,int x2,int y2)
{
float x,y,xinc,yinc,dx,dy;
int k;
int step;
dx=x2-x1;
dy=y2-y1;
if(abs(dx)>abs(dy))
step=abs(dx);
else
step=abs(dy);
```

```
xinc=dx/step;
yinc=dy/step;
x=x1;
y=y1;
putpixel(x,y,1);
for(k=1;k<=step;k++)
{
x=x+xinc;
y=y+yinc;
putpixel(x,y,2);
}
}
```

Above program draws a line from (100, 100) to (200, 200).

line program.

Output of program:



Sample Viva Question (Experiment No.4)

1. **What is DDA?**
2. **What is run length encoding?**
3. **What is the need of homogeneous coordinates?**
4. **What are the disadvantages of DDA algorithm?**
5. **What is Translation?**

****

Experiment No -5

Aim:-Write a program to implement midpoint circle algorithm.

Source Code:-

```
#include<stdio.h>
#include<conio.h>
#include<graphics.h>
void main()
{

 int gd=DETECT,gm;
 int x,y,r;
 void Drawcircle(int,int,int);
 printf("Enter the Mid points and Radious:");
 scanf("%d%d%d",&x,&y,&r);
 initgraph(&gd,&gm,"");
 Drawcircle(x,y,r);
 getch();
 closegraph();

}
void Drawcircle(int x1,int y1,int r)
{
 int x=0,y=r,p=1-r;
 void cliplot(int,int,int,int);
 cliplot(x1,y1,x,y);
 while(x<y)
 {
 x++;
 if(p<0)
  p+=2*x+1;
  else
  {
   y--;
   p+=2*(x-y)+1;
  }
```

```
 cliplot(x1,y1,x,y);
 }

}
void cliplot(int xctr,int yctr,int x,int y)
{

 putpixel(xctr +x,yctr +y,1);
 putpixel(xctr -x,yctr +y,1);
 putpixel(xctr +x,yctr -y,1);
 putpixel(xctr -x,yctr -y,1);
 putpixel(xctr +y,yctr +x,1);
 putpixel(xctr -y,yctr +x,1);
 putpixel(xctr +y,yctr -x,1);
 putpixel(xctr -y,yctr -x,1);
 getch();

}
```

Output of program:



Sample Viva Question (Experiment No -5)

1. **Define Circle.**
2. **How many memory bits are required for a 24-bit plane 1024 × 1024 element raster?**
3. **What is the Rotation?**
4. **Distinguish between uniform scaling and differential scaling?**
5. **What is Back face removal?**

****

Experiment No - 6

Aim:-Write a program to fill an ellipse.

Theory:-

Source Code:-

```
#include <graphics.h>

#include <stdlib.h>

#include <stdio.h>

#include <conio.h>


int main(void)

{

 /* request autodetection */

 int gdriver = DETECT, gmode, errorcode;

 int midx, midy, i;

 int xradius = 100, yradius = 50;


 /* initialize graphics and local variables */

 initgraph(&gdriver, &gmode, "c:\\tc\\bgi");


 /* read result of initialization */

 errorcode = graphresult();

 if (errorcode != grOk) {  /* an error occurred */

  printf("Graphics error: %s\n", grapherrormsg(errorcode));

  printf("Press any key to halt:");

  getch();
```

```
  exit(1);          /* terminate with an error code */
 }

 midx = getmaxx() / 2;
 midy = getmaxy() / 2;

 /* loop through the fill patterns */
 for (i = EMPTY_FILL; i < USER_FILL; i++) {
  /* set fill pattern */
  setfillstyle(i, COLORS(12));

  /* draw a filled ellipse */
  fillellipse(midx, midy, xradius, yradius);
  getch();
 }

 /* clean up */
 closegraph();
 return 0;
}
```

Output of program:



Sample Viva Question (Experiment No -6)

1. **What is Mid Point Ellipse Algorithm?**
2. **Define Ellipse.**
3. **What is Boundary Fill Algorithm?**
4. **What is Flood Fill/ Seed Fill Algorithm?**
5. **Difference between Boundary Fill & Flood Fill Algorithm.**

****

Experiment No - 7

Aim:- Write a program to fill a polygon.

Theory:-

Source Code:-

```
#include <graphics.h>
#include <stdlib.h>
#include <stdio.h>
#include <conio.h>

int main(void)
{
  int gdriver = DETECT, gmode, errorcode;
  int i, x, y;
  int corner[8];
  initgraph(&gdriver, &gmode, "c:\\tc\\bgi");
  errorcode = graphresult();
  if (errorcode != grOk)
  {
    printf("Graphics error: %s\n", grapherrormsg(errorcode));
    printf("Press any key to halt:");
    getch();
    exit(1);
  }
  x = getmaxx()/4;
```

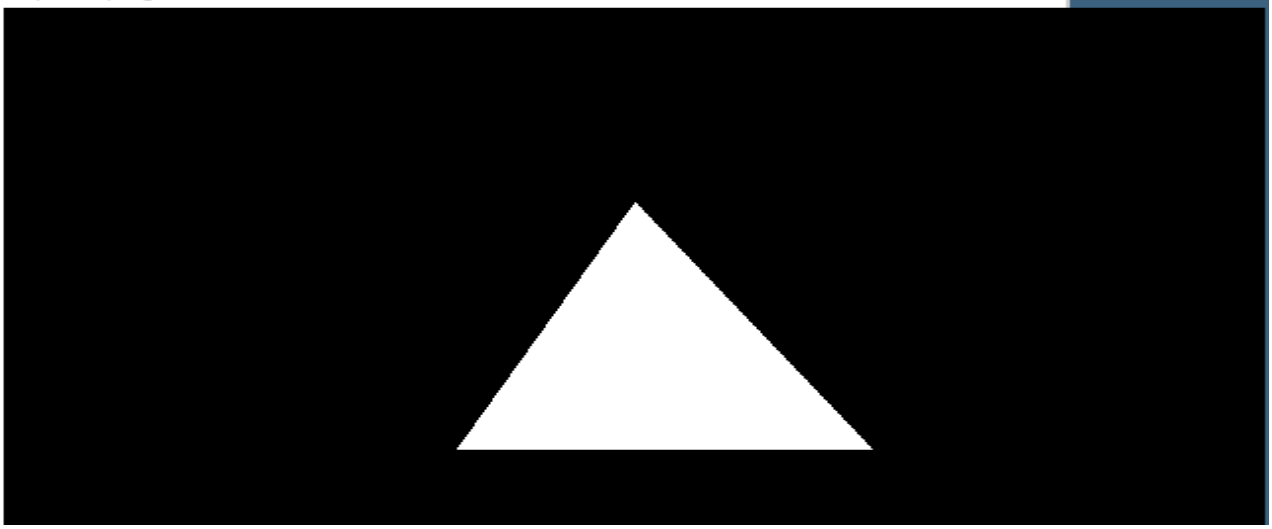
```
  y = getmaxy()/4;


  corner[0] = x;
  corner[1] = y;


  corner[2] = x+200;
  corner[3] = y+0;


  corner[4] = x+200;
  corner[5] = y+150;
 corner[6] = x+0;
  corner[7] = y+150;
  for (i=EMPTY_FILL; i<USER_FILL; i++)
  {
    setfillstyle(i, COLORS(random(15)));
    fillpoly(4, corner);
    getch();
  }
  closegraph();
  return 0;
}
```

Output of program:



Sample Viva Question (Experiment No -7)

1. **Explain the illumination model.**
2. **What is a spline?**
3. **Define B-Spline curve.**
4. **What are the different ways of specifying spline curve?**
5. **What are the important properties of Bezier Curve?**

****

Experiment No - 8

Aim:- Write a program to hoist a flag.

Source Code:-

```
#include <dos.h>

#include <graphics.h>

#include <stdlib.h>

#include <stdio.h>

#include <conio.h>


int main(void)

{


  int gdriver = DETECT, gmode,poly[10],poly1[10],poly2[10],poly3[10];

  initgraph(&gdriver, &gmode, "c:\\tc\\bgi");


//pole

   {

        poly[0]=50;

        poly[1]=50;


        poly[2]=60;

        poly[3]=50;


        poly[4]=60;

        poly[5]=400;
```

```
poly[6]=50;

poly[7]=400;


poly[8]=50;

poly[9]=50;


setfillstyle(SOLID_FILL,WHITE);

 fillpoly(4,poly);

      }



//          flag

      {

poly1[0]=60;

poly1[1]=50;


poly1[2]=320;

poly1[3]=50;


poly1[4]=320;

poly1[5]=100;


poly1[6]=60;

poly1[7]=100;
```

```
poly1[8]=60;

poly1[9]=50;


setfillstyle(SOLID_FILL,RED);

 fillpoly(4,poly1);

      }

      {


poly2[0]=60;

poly2[1]=100;


poly2[2]=320;

poly2[3]=100;


poly2[4]=320;

poly2[5]=150;


poly2[6]=60;

poly2[7]=150;


poly2[8]=60;

poly2[9]=100;


setfillstyle(SOLID_FILL,WHITE);
```

```
 fillpoly(4,poly2);


}
      {
poly3[0]=60;
poly3[1]=50;
poly3[2]=320;
poly3[3]=50;
poly3[4]=320;
poly3[5]=200;
poly3[6]=60;
poly3[7]=200;
poly3[8]=60;
poly3[9]=50;
setfillstyle(SOLID_FILL,RED);
 fillpoly(4,poly3);
   }
{
 setfillstyle(SOLID_FILL,BLINK);
 pieslice(190,126 ,0,45,25);
 pieslice(190,126 ,45,90,25);
 pieslice(190,126 ,90,135,25);
 pieslice(190,126 ,135,180,25);
 pieslice(190,126 ,180,225,25);
 pieslice(190,126 ,225,270,25);
```

```
        pieslice(190,126 ,270,315,25);

        pieslice(190,126 ,315,360,25);

        }

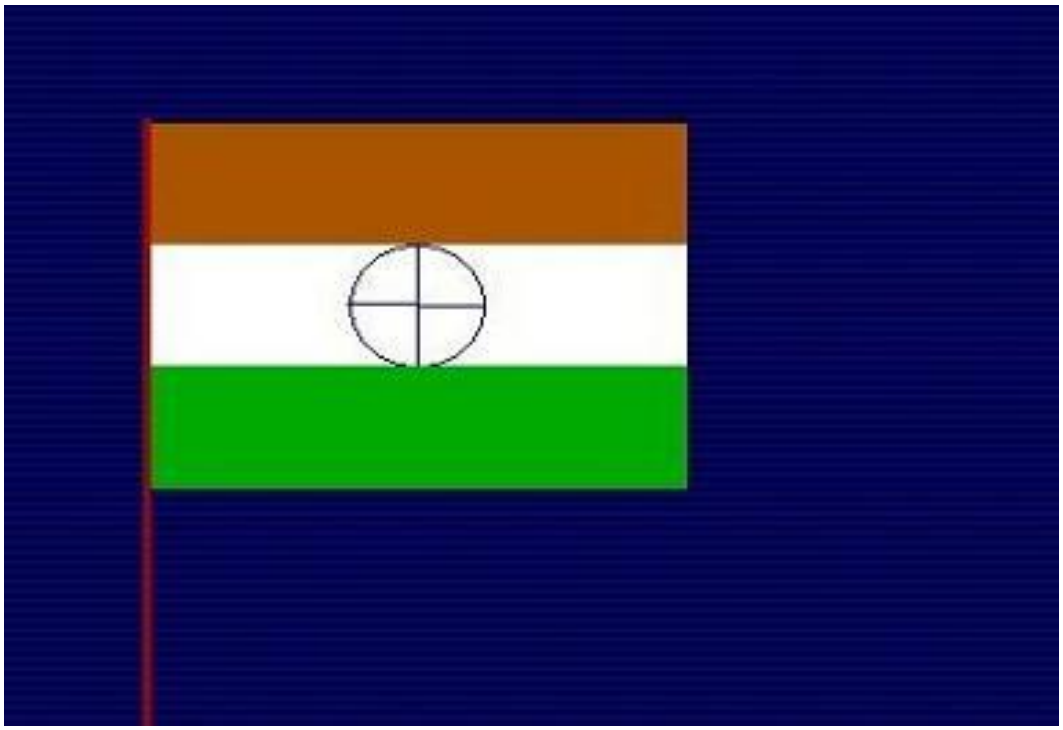
getch();

  closegraph();

  return 0;

}
```

Output:-



Sample Viva Question (Experiment No -8)

1. **What are the design issues for multimedia authoring?**
2. **What is panning?**
3. **Define mail message.**
4. **What are the components of a distributed Multimedia system?**
5. **What are the primary n/w topologies used for multimedia?**

****

Experiment No - 9

Aim:- Write a program to create balloons.

Theory:-

Source Code:-

```
#include <dos.h>

#include <graphics.h>

#include <stdlib.h>

#include <stdio.h>

#include <conio.h>


int main(void)

{


  int gdriver = DETECT, gmode, errorcode;


   initgraph(&gdriver, &gmode, "c:\\tc\\bgi");

    while(!kbhit())

    {

     int i,j=0;

    for(i=440,j=0;i>20,j<400;i--,j++)

    {

       int k=180+j;


   setfillstyle (SOLID_FILL,WHITE);
```

```
fillellipse(k,i,50,90);
fillellipse(k,i+95,5,10);
line(k+5,i+100,k-20,i+270);
setfillstyle (SOLID_FILL,RED);
fillellipse(180,i,50,90);
fillellipse(180,i+95,5,10);
line(185,i+100,220,i+300);
setfillstyle (SOLID_FILL,BLUE);
fillellipse(250,i,50,90);
fillellipse(250,i+95,5,10);
line(250,i+100,220,i+300);
setfillstyle (SOLID_FILL,GREEN);
fillellipse(310,i,50,90);
fillellipse(310,i+95,5,10);
line(310,i+100,220,i+300);
setfillstyle (SOLID_FILL,MAGENTA);
fillellipse(210,i+60,50,90);
fillellipse(210,i+155,5,10);
line(210,i+160,220,i+300);
setfillstyle (SOLID_FILL,YELLOW);
fillellipse(270,i+60,50,90);
fillellipse(270,i+155,5,10);
line(270,i+160,220,i+300);
 delay(50);
 cleardevice();
```

```
    }
   }
   getch();
   closegraph();
   return 0;
}
```

Output:-



Sample Viva Question (Experiment No -9)

1. **List different color models.**
2. **Different between database and multimedia database.**
3. **List out the input devices of multimedia.**
4. **What is the use of control points?**
5. **Define Projection.**

****

Experiment No -10

Aim:-Write a program to implement a walking robot.

Source Code:-

```
#include<dos.h>
#include <graphics.h>
#include <stdlib.h>
#include <stdio.h>
#include <conio.h>

int main(void)
{
  int gdriver = DETECT, gmode, errorcode;
  initgraph(&gdriver, &gmode, "c:\\tc\\bgi");
  errorcode = graphresult();
  if (errorcode != grOk)
  {
    printf("Graphics error: %s\n", grapherrormsg(errorcode));
    printf("Press any key to halt:");
    getch();
    exit(1);
  }
```

```
int i,x,y,poly[10];
{
while(!kbhit() && (x=620))
{
for(x=0;x<=620;x=x+10)
{
/*f*/

setfillstyle(SOLID_FILL,COLORS(6));
fillellipse(x+50,150,25,25);
setfillstyle(SOLID_FILL,COLORS(BLACK));
fillellipse(x+65,145,3,3);

/*c*/
 poly[0]=x+25;
 poly[1]=175;

 poly[2]=x+75;
 poly[3]=175;

 poly[4]=x+75;
 poly[5]=300;

 poly[6]=x+25;
 poly[7]=300;
```

```
poly[8]=x+25;
poly[9]=175;

drawpoly(5,poly);
/*h1*/
line(x+50,200,x+20,315);
line(x+52,202,x+20,320);
/*h2*/
line(x+75,230,x+90,280);
line(x+75,232,x+94,285);
        /*l1*/
line(x+50,300,x+100,430);
line(x+57,300,x+107,430);
        /*l2*/
line(x+46,300,x+25,430);
line(x+53,300,x+32,430);

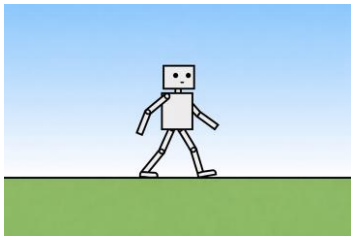
                /*p1*/
line(x+23,430,x+55,430);
line(x+23,435,x+55,435);
            /*p2*/

line(x+97,430,x+129,430);
line(x+97,435,x+129,435);
```

```
delay(150);
cleardevice();
  }


  }
}
   getch();
   closegraph();
   return 0;
}
```

Output:



Sample Viva Question (Experiment No -10)

1. **What is JPEG?**
2. **Assuming that a certain full-color (24 bit per pixel) RGB raster system has a 512 by 512 frame buffer, how many distinct color choices (intensity levels) would be available?**
3. **What is the difference between impact and non-impact printers?**
4. **What is projector?**
5. **What is hypermedia documents?**

****

Beyond The Syllabus

Experiment No - 1

Aim:- Write a program to implement a rotating wheel.

Source Code:-

```
#include<graphics.h>
#include<dos.h>
#include<conio.h>
#include<stdio.h>
#include<stdlib.h>
int main(void)
{
int gdriver = DETECT, gmode, errorcode;
int i,x,y;
initgraph(&gdriver, &gmode, "c:\\tc\\bgi”);
errorcode = graphresult();
if (errorcode != grOk)
{
  printf("Graphics error: %s\n", grapherrormsg(errorcode));
  printf("Press any key to halt:");
  getch();
  exit(1);
}
int j;
x=0;
```

```
y=getmaxy()/2;
while (!kbhit() && (x=640))
{
   for(x=0;x<=620;x=x+15)
   {
   circle(x,y,100);
    {
    circle(x,y,90);
   pieslice(x,y,0, 90, 90);
   pieslice(x,y,90,180, 90);
   pieslice(x,y,180,270, 90);
   pieslice(x,y,270,360, 90);
    delay(150);
   circle(x,y,5);
   cleardevice();
 }
     circle(x,y,100);
   circle(x,y,90);
   pieslice(x,y,0, 45, 90);
   pieslice(x,y,90,135, 90);
   pieslice(x,y,180,225, 90);
   pieslice(x,y,270,315, 90);
  setfillstyle(EMPTY_FILL,getmaxcolor());
   circle(x,y,5);
   delay(150);
```

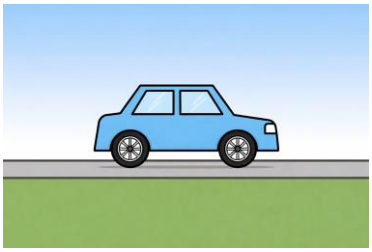
```
    cleardevice();

}}

closegraph();

exit(1);}
```

Output:



Sample Viva Question (Experiment No -1)

1. **What is rotation in computer graphics?**
2. **Write the 2D rotation transformation matrix.**
3. **How do you perform rotation about an arbitrary point (x, y)?**
4. **What is the difference between clockwise and anticlockwise rotation?**
5. **What is the difference between 2D and 3D rotation?**

****

Experiment No - 2

Aim:- Write a program to implement Cohen Sutherland algorithm.

Source Code:-

```
#include<conio.h>
#include<stdio.h>
#include<graphics.h>
#include<math.h>

typedef unsigned char BYTE; // 0 to 255 or 0 to 1111-1111 ie 0xff
BYTE pointcheck(int x, int y);
BYTE bit1check(int p);
BYTE bit2check(int p);
BYTE bit3check(int p);
BYTE bit4check(int p);
int endprog(void);
int main(void)
{
//graphics mode initialzation to 640x480 resolution
int gd=VGA,gm=VGAHI;
initgraph(&gd,&gm,"");
//window defined
float x,y,xmin,ymin,xmax,ymax;
xmax=540,ymax=380,xmin=100;ymin=100;
//line end points defined
float xn1,xn2,yn1,yn2,x1,y1,x2,y2;
```

```
int p1,p2;
x1=120;y1=12;x2=300;y2=390; // with less values of point 1
//draw initial situtation
setcolor(1);
rectangle(xmin,ymin,xmax,ymax);
setcolor(2);
line(x1,y1,x2,y2);
//program of clipping begins here
//bits are numbered as 1    1    1      1
//              ^fourth ^third ^second ^first
float m=(y2-y1)/(x2-x1);  //slope calculated
//getting binary conversion of points
p1=pointcheck(x1,y1);
p2=pointcheck(x2,y2);
//printing line position
int bit1,bit2,bit3,bit4;
bit1=bit1check(p1);
bit2=bit2check(p1);
bit3=bit3check(p1);
bit4=bit4check(p1);
printf("point p1 is %d =%d%d%d%d",p1,bit4,bit3,bit2,bit1);
bit1=bit1check(p2);
bit2=bit2check(p2);
bit3=bit3check(p2);
bit4=bit4check(p2);
```

```
printf("\n point p2 is %d=%d%d%d%d",p2,bit4,bit3,bit2,bit1);

//if the line lies outside exit program

if (bit1check(p1)&bit1check(p2)==1)endprog();

if (bit2check(p1)&bit2check(p2)==1)endprog();

if (bit3check(p1)&bit3check(p2)==1)endprog();

if (bit4check(p1)&bit4check(p2)==1)endprog();

if (p1==0 & p2==0)

{

printf("\n line lies within the window");

getch();

return 0;

}

else printf("\n clipping ....");

//clipping required from interesting points .calculating that..

int pointcount=0;

if(pointcheck(x1,y1)==0)  //if it starts from within window

{

xn1 = x1;yn1 = y1;

pointcount =pointcount +1;

p1=p2;

}

if(bit1check(p1)==1) //it crosses xmin

{

x=xmin;

y=y1+m*(xmin-x1);
```

```
putpixel(x,y,3);
setcolor(3);
circle(x,y,4);
if(pointcount>0)
{
xn2=x;yn2=y;
}
else {
xn1=x;yn1=y;
pointcount=pointcount+1;
}
p1=p2;
}
if(bit2check(p1)==1)//it crosses xmax
{
x=xmax;
y=y1+m*(xmax-x1);
putpixel(x,y,3);
setcolor(3);
circle(x,y,4);
if(pointcount>0)
{
xn2=x;yn2=y;
}
else {
```

```
xn1=x;yn1=y;
pointcount=pointcount+1;
}
p1=p2;
}
if(bit3check(p1)==1)  //it crosses ymin
{
y=ymin;
x=x1+(ymin-y1)/m;
putpixel(x,y,3);
setcolor(3);
circle(x,y,4);
if (pointcount >0)
{
xn2=x;yn2=y;
}
else {
xn1=x;yn1=y;
pointcount=pointcount+1;
}
p1=p2;
}
if(bit4check(p1)==1)  //it crosses ymax
{
y=ymax;
```

```
x=x1+(ymax-y1)/m;
putpixel(x,y,3);
setcolor(3);
circle(x,y,4);
if (pointcount>0)
{
xn2=x; yn2=y;
}
else {
xn1=x; yn1=y;
pointcount=pointcount+1;
}
}
getch();
setcolor(0);
line(x1,y1,x2,y2);
setcolor(4);
line(xn1,yn1,xn2,yn2);
getch();
closegraph();
}
int endprog(void)
{
printf("\n line lies outside the window");
getch();
```

```
closegraph();
return 0;
}
//function returns binary value of point location
BYTE pointcheck(int x,int y)
{
//xmax=540;ymax=380;xmin=100;ymin=100;
int val=0x00;
if((x-100)<0) val=val|0x01;//make bit 1-->1
if((540-x)<0) val=val|0x02; //make bit 2 -->1
if((y-100)<0) val=val|0x04; //make bit 3 -->1
if((380-y)<0) val=val|0x08; //make bit 4 -->1
return val;
}
//subroutines for checking individual bit values
BYTE bit1check(int a)
{
a=a&1;  //and operation with 0001,making 2,3,4 bit0
return a;
}
BYTE bit2check(int b)
{
b=b&2;//and with 0010 i.e making 1,3,4,bit 0
b=b>>1;  //right shift by 1 bit i.e cutting last digit
return b;
```

```
}

BYTE bit3check(int c)

{

c=c&4;  //and with 0100 i.e making 1,2,4 bit 0

c=c>>2;  //right shift 2 digits i.e cutting last two digit

return c;

}

BYTE bit4check(int d)

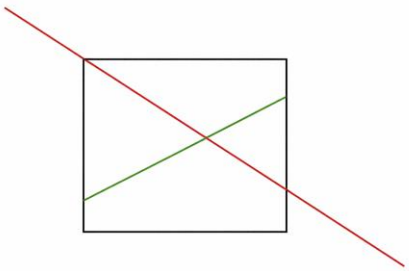
{

d=d>>3;  //right shift 3 digits i.e cutting last three digit

return d;

}
```

Output:-



Sample Viva Question (Experiment No -2)

1. **What is the Cohen–Sutherland algorithm?**
2. **What is an out code in the Cohen–Sutherland algorithm?**
3. **What are trivial acceptance and trivial rejection?**
4. **How does the algorithm calculate intersection points?**
5. **What are the advantages of the Cohen–Sutherland algorithm?**

****

Viva Questions

Q1. What is Pixel?

Answer:- A Pixel (Picture Element) is the smallest unit of a digital image or display screen. It represents a single point of color or intensity.

An image is made up of a grid of thousands or millions of pixels.

Q2. What is Bitmap?

Answer:- A Bitmap is a representation of an image as a matrix of pixels, where each pixel corresponds to a specific location and color value. It is also known as a raster image. In a bitmap, the image is stored as a collection of bits in memory, where each bit or group of bits represents a pixel. The quality of a bitmap image depends on its resolution and color depth. Common bitmap formats include BMP, JPEG, PNG, etc. Bitmap images require more storage space as resolution increases. They are widely used in digital imaging and graphics applications.

Q3. What is Refresh Buffer / Frame Buffer?

Answer:- A Refresh Buffer (Frame Buffer) is a dedicated area in memory used to store pixel data for the display screen. It holds intensity or color values for every pixel on the screen. The display controller continuously reads data from the frame buffer and refreshes the screen at regular intervals, typically 60 times per second. This process ensures that the image remains stable and visible. Frame buffers are essential in raster scan display systems. The size of the buffer depends on screen resolution and color depth.

Q4. How many Bits are used in Bitmap?

Answer:- The number of bits used in a bitmap image depends on the color depth (bits per pixel). It determines how many colors can be represented. For example, 1-bit represents black and white images, 8-bit allows 256 colors, 16-bit supports 65,536 colors, and 24-bit (true color) supports around 16.7 million colors. Some systems use 32-bit, which includes an alpha channel for transparency. Higher bits per pixel result in better image quality but require more memory. Thus, bitmap size is directly proportional to resolution and color depth.

Q5. What is Pixmap?

Answer:- A Pixmap is an extension of bitmap that stores detailed color information for each pixel. Unlike bitmap, which may store only binary data (0 or 1), pixmap represents multiple colors using several bits per pixel. It is used in advanced graphics systems to handle colored images efficiently. Pixmaps allow high-quality image representation with smooth color variations. They are commonly used in graphical user interfaces (GUI) and modern display systems. Essentially, all colored images in computers are stored as pixmaps.

Q6. What is Clipping?

Answer: Clipping is a technique used in computer graphics to remove the parts of objects that lie outside a specified boundary called the **clipping window**.

Only the portion of the object inside the window is displayed.

It helps improve efficiency by avoiding unnecessary drawing.

Clipping can be applied to lines, polygons, text, and curves.

Example:
If a line extends outside the screen, only the visible part inside the screen is shown.

Q7. Difference between Raster Scan and Random Scan

Answer:

Basis	**Raster Scan System**	**Random Scan System**
Definition	Scans entire screen line by line	Draws only required lines
Representation	Pixel-based	Vector-based
Refreshing	Whole screen refreshed	Only drawn objects refreshed
Quality	Lower resolution	High-quality smooth lines
Cost	Low	High
Application	TV, Monitors	CAD systems, Oscilloscopes

Q8. Bits required for a 512 × 512 raster with 3 bits per pixel

Answer:

Given:
Resolution = 512 × 512
Bits per pixel = 3

Formula:

Total Bits=Resolution×Bits per Pixel

Calculation:

512×512=262144 pixels

262144×3=786432 bits

Final Answer:
Total bits required = 786,432 bits

Q9. Define Point Clipping

Answer:
Point clipping is the process of determining whether a given point lies inside or outside a clipping window.

If the point is inside → it is displayed.

If the point is outside → it is discarded.

Condition:
If a point (x,y) satisfies:

$x_{min} \leq x \leq x_{max}$,

$y_{min} \leq y \leq y_{max}$

then it lies inside the window.

Q10. Pixels accessed per second and access time per pixel

Answer:

Given:
Resolution = 2560 × 2048
Refresh rate = 60 frames/sec

Step 1: Total pixels per frame

2560×2048=5,242,880 pixels

Step 2: Pixels accessed per second

5,242,880×60=314,572,800 pixels/sec

Step 3: Access time per pixel

Access Time=1/314,572,8001

$\approx 3.18 \times 10^{-9}$ seconds

Final Answer:

Pixels accessed per second = **314,572,800 pixels/sec**

Access time per pixel = **3.18 ns**

Q11. Advantages of Bresenham's Algorithm .

Answer:

Advantages:

- Uses only integer arithmetic (no floating-point operations).
- Faster execution compared to DDA algorithm.
- Reduces round-off errors.
- Efficient for real-time applications.
- Simple and suitable for hardware implementation.

Q12. Path of electron beam when returning to left side of CRT screen .

Answer:

The path taken by the electron beam when it returns to the left side of the screen after completing one horizontal line is called Horizontal Retrace (or Horizontal Flyback).

During this time, the beam is turned off (blanked).

It prepares to draw the next line.

Q13. What is the refresh rate of Flicker?

Answer:

Flicker occurs when the refresh rate of a display is too low.

If refresh rate is below 50–60 Hz, flickering becomes noticeable.

To avoid flicker, the refresh rate should be at least 60 Hz or higher.

Conclusion:

Flicker is noticeable typically below 60 Hz refresh rate.

Q14. Short notes on Active and Passive Transformations .

Answer:

1. Active Transformation:

- The object itself is moved or transformed.
- Coordinate system remains fixed.
- Example: Moving a triangle from one position to another.

2. Passive Transformation:

- The coordinate system is changed while the object remains fixed.
- Equivalent visual effect as active transformation.

- Example: Changing the viewing position.

Difference:

Active → Object moves

Passive → Coordinate system moves

Q15. Path of electron beam at the end of each refresh cycle .

Answer:

The path taken by the electron beam at the end of a complete screen refresh is called Vertical Retrace (or Vertical Flyback).

The beam returns from the bottom-right corner to the top-left corner.

The beam is blanked during this time.

It prepares for the next frame.

Q16. What is DDA? (Digital Differential Analyzer)

Answer:

DDA (Digital Differential Analyzer) is a **line drawing algorithm** used in computer graphics to draw a straight line between two points.

- It works by calculating intermediate points using **incremental steps**.
- Based on slope $m=\Delta y\Delta x$
- Coordinates are updated step-by-step and rounded to nearest integer.

Algorithm Steps (brief):

1. Calculate $dx=X_2-X_1$ $dy=Y_2-Y_1$
2. Steps = max(|dx|, |dy|)
3. Increment: xinc=dx/stepsx_{inc} = dx/stepsxinc=dx/steps, yinc=dy/stepsy_{inc} = dy/stepsyinc=dy/steps
4. Plot points until end point is reached

Q17. What is Run Length Encoding (RLE)? *(5 Marks)*

Answer:
Run Length Encoding (RLE) is a **lossless data compression technique**.

- It compresses data by replacing repeated values with a **count and value**.
- Commonly used in **bitmap images**.

Example:
AAAABB → (4A, 2B)

Advantages:

- Simple and easy to implement
- Effective for images with large uniform areas

Q18. What is the need of Homogeneous Coordinates? ***(5 Marks)***

Answer:
Homogeneous coordinates are used in computer graphics to represent transformations in a unified mathematical form.

Need/Advantages:

1. Allows **translation** to be represented as matrix multiplication.
2. Combines multiple transformations (scaling, rotation, translation) into a **single matrix**.
3. Simplifies mathematical calculations.
4. Enables use of **3×3 matrices (2D)** and **4×4 matrices (3D)**.
5. Useful in computer graphics pipelines.

Q19. Disadvantages of DDA Algorithm ***(5 Marks)***

Answer:
Although DDA is simple, it has some limitations:

1. Uses **floating-point arithmetic** → slower than integer methods.
2. Requires **rounding**, leading to inaccuracies.
3. Accumulation of **round-off errors**.
4. Less efficient compared to Bresenham's algorithm.
5. Not suitable for high-performance graphics systems.

Q20. What is Translation? ***(5 Marks)***

Answer:
Translation is a basic transformation in computer graphics that **moves an object from one position to another** without changing its shape or size.

In 2D:
If a point (x,y) is translated by (tx,ty):

$$x' = x + tx, y' = y + ty$$

- Object shifts in **x and/or y direction**
- Shape, size, and orientation remain unchanged
- Represented using homogeneous coordinates in matrix form

Q21. Define Circle

Answer:
A **circle** is a set of all points in a plane that are at a constant distance (radius) from a fixed point called the **center**.

Mathematical Equation:

$(x-h)^2+(y-k)^2=r^2$

Where:

- (h,k) = center of the circle
- r = radius

Key Points:

- All points are equidistant from the center
- Widely used in computer graphics for drawing curves

Q22. Memory bits required for 24-bit plane (1024 × 1024 raster) .

Answer:

Given:
Resolution = 1024 × 1024
Bits per pixel = 24

Formula:

Total Memory=Resolution×Bits per Pixel

Calculation:

1024×1024=1,048,576 pixels

1,048,576×24=25,165,824 bits

Final Answer:
Required memory = 25,165,824 bits

Q23. What is Rotation?

Answer:
Rotation is a transformation that **turns an object about a fixed point (pivot)** through a specified angle.

In 2D Rotation:
If a point (x,y) is rotated by angle θ about the origin:

$x' = x\cos\theta - y\sin\theta$

$y' = x\sin\theta + y\cos\theta$

Key Points:

- Rotation can be clockwise or anticlockwise
- Shape and size remain unchanged
- Orientation of object changes

Q24. Difference between Uniform Scaling and Differential Scaling.

Basis	Uniform Scaling	Differential Scaling
Definition	Same scaling factor in all directions	Different scaling factors in different directions
Scale Factors	(S_x = S_y)	(S_x \neq S_y)
Shape	Shape preserved	Shape may distort
Example	Enlarging a square equally	Stretching rectangle in one direction

Conclusion:

- Uniform scaling maintains proportions
- Differential scaling changes proportions

Q25. What is Back Face Removal?

Answer:
Back face removal (or back face culling) is a technique used in 3D graphics to **eliminate surfaces of objects that are not visible to the viewer**.

Key Points:

- Only front faces (visible surfaces) are rendered
- Back faces are those facing away from the viewer
- Improves rendering efficiency

Advantage:

- Reduces computation time and improves performance

Q26. What is Mid-Point Ellipse Algorithm?

Answer:
The **Mid-Point Ellipse Algorithm** is used to draw an ellipse efficiently in computer graphics using **decision parameters**.

- It determines the next pixel position based on the midpoint between two possible pixels.
- Uses **only integer calculations**, making it efficient.
- The ellipse is divided into **two regions**:

1. Region 1: slope > –1
2. Region 2: slope < –1

Steps (brief):

1. Start from point (0,b)(0, b)(0,b)
2. Calculate decision parameter
3. Choose next pixel based on midpoint
4. Use symmetry to draw all four quadrants

Q27. Define Ellipse .

Answer:
An **ellipse** is the set of all points in a plane such that the **sum of distances from two fixed points (foci)** is constant.

Standard Equation:

$$X^2/a^2+Y^2/b^2=1$$

Where:

- a = semi-major axis
- b = semi-minor axis

Key Points:

- Special case of a circle when a=b
- Used in computer graphics for curved shapes

Q28. What is Boundary Fill Algorithm?

Answer:
Boundary Fill Algorithm is a **region filling technique** used to fill an area bounded by a specific boundary color.

- Starts from a **seed point inside the region**
- Fills pixels until it reaches the boundary color

Types:

- 4-connected
- 8-connected

Condition:
Fill continues until boundary color is encountered.

Q29. What is Flood Fill (Seed Fill) Algorithm?

Answer:
Flood Fill Algorithm is a **region filling method** that fills an area with a new color by replacing all connected pixels of the same color.

- Starts from a **seed point**
- Replaces all pixels having the same **old color**

Key Points:

- Works on **uniform regions**
- Widely used in paint programs

Q30. Difference between Boundary Fill and Flood Fill Algorithm .

Answer:

Basis	Boundary Fill	Flood Fill
Filling Criteria	Stops at boundary color	Replaces interior color
Requirement	Needs boundary color	Needs target (old) color
Working	Fills until boundary reached	Fills all similar pixels
Application	Regions with clear boundary	Regions with uniform color
Example	Coloring shapes with borders	Paint bucket tool

Conclusion:

- Boundary Fill → boundary-based
- Flood Fill → color-based

Q31. Explain the Illumination Model .

Answer:

An **illumination model** (or lighting model) is a mathematical method used in computer graphics to simulate how light interacts with surfaces to produce realistic images.

It determines the color and brightness of objects based on light sources, surface properties, and viewer position.

Components:

- **Ambient Reflection** – General background light (soft, uniform light)
- **Diffuse Reflection** – Light scattered equally in all directions (depends on angle)
- **Specular Reflection** – Shiny highlights (depends on viewer direction)

Q32. What is a Spline?

Answer:
A **spline** is a smooth curve defined using a set of **control points**.

- Used to model complex shapes smoothly
- Widely used in **computer graphics, CAD, animation**

Key Features:

- Smooth and continuous curve
- Controlled by control points
- Can approximate or interpolate points

Q33. Define B-Spline Curve.

Answer:- A **B-Spline (Basis Spline) Curve** is a generalization of Bezier curves that provides greater control and flexibility.

Definition:
A B-Spline curve is a piecewise polynomial curve defined by a set of control points and basis functions.

Key features:

- Local control over shape
- Smooth and continuous
- Does not necessarily pass through control points

Q34. Different Ways of Specifying a Spline Curve

Answer:- Spline curves can be specified in different ways:

A. **Control Points Method**

Curve defined using a set of control points

B. **Polynomial Representation**

Using mathematical equations (cubic, quadratic)

C. **Blending Functions**

Curve formed using blending (basis) functions

D. **Parametric Form**

Represented as functions of a parameter (t)

E. **Interpolation Method**

Curve passes through given data points

Q35. Important Properties of Bezier Curve

Answer:- A **Bezier curve** is defined using control points and Bernstein polynomials.

Properties:

1. **Endpoint Property**

Curve passes through first and last control points

2. **Convex Hull Property**

   Curve lies inside the convex hull of its control points

3. **Variation Diminishing Property**

   Curve does not oscillate more than control polygon

4. **Global Control**

   Changing one control point affects entire curve

5. **Affine Invariance**

   Transformations (translation, scaling, rotation) apply uniformly

6. **Continuity**

   Smooth and continuous curve

Q36. What are the design issues for multimedia authoring?

Answer:- The major design issues involved in multimedia authoring are:

- User Interface Design – Should be simple, interactive, and user-friendly.
- Synchronization – Proper timing between audio, video, and animation.
- Media Integration – Combining text, graphics, audio, video smoothly.
- Navigation Structure – Easy movement through content (menus, links).
- Performance & Storage – Efficient use of memory and fast loading.
- Scalability – Ability to handle large multimedia content.
- Interactivity – User control over playback and content flow.

Q37. What is Panning?

Answer:- Panning is a technique used in multimedia (especially audio and video) to create the effect of movement across the screen or sound field.

- In audio, it refers to shifting sound from left to right speakers.
- In video/graphics, it means moving the camera view horizontally across a scene.

Q38. Define Mail Message.

Answer:- A mail message (email message) is a digital communication sent over a network.

It typically consists of:

- Header (sender, receiver, subject, date)
- Body (main content/message)
- Attachments (optional files like images, documents, etc.)

Q39. What are the components of a distributed Multimedia system?

Answer:- A distributed multimedia system consists of:

- **Multimedia Servers** – Store and deliver multimedia data.
- **Client Systems** – End-user devices that request and display content.
- **Network Infrastructure** – Communication medium (Internet, LAN, etc.).
- **Database Management System (DBMS)** – Manages multimedia data.
- **Middleware** – Handles communication between distributed components.
- **Compression/Decompression Units** – Efficient data transmission.

Q40. What are the primary n/w topologies used for multimedia?
Answer:- The **primary network topologies used for multimedia systems** are:

A. Star Topology

- All devices are connected to a central hub or switch.
- Most commonly used in multimedia networks.
- Advantages: Easy management, fault isolation.
- Limitation: Failure of the central hub affects the entire network.

B. Mesh Topology

- Every device is connected to every other device.
- Provides high reliability and redundancy.
- Suitable for high-performance multimedia applications.
- Limitation: Expensive and complex.

C. Hybrid Topology

- Combination of two or more topologies (e.g., star + bus).
- Flexible and scalable.
- Widely used in modern multimedia systems.

Q41. List different color models

Answer:- Common color models used in multimedia are:

- **RGB (Red, Green, Blue) – Used in displays (monitors, TVs)**
- **CMY/CMYK (Cyan, Magenta, Yellow, Black) – Used in printing**
- **HSV (Hue, Saturation, Value) – Used in graphics and image editing**
- **HSL (Hue, Saturation, Lightness)**
- **YUV / YCbCr – Used in video and television systems**

Q42. Difference between database and multimedia database
Answer:-

Basis	**Database**	**Multimedia Database**
Data Type	Stores structured data (text, numbers)	Stores multimedia data (images, audio, video)
Complexity	Simple	Complex due to large file sizes
Storage	Less storage required	Requires large storage
Retrieval	Based on queries (SQL)	Content-based retrieval (image/video search)
Example	Student records system	Video streaming system

Q43. List out the input devices of multimedia

Answer:- Keyboard ,Mouse ,Scanner ,Microphone ,Digital Camera ,Webcam ,Touch Screen ,Graphics Tablet ,Joystick

Q44. What is the use of control points?
Answer:-

- Control points are used in graphics and animation to define and shape curves and surfaces.
- They help in designing objects like splines and Bézier curves.
- By moving control points, the shape of the curve or object can be easily modified.

Q45. Define Projection
Answer:-

- Projection is the process of converting 3D objects into 2D representations on a display screen.
- It is widely used in computer graphics.
- Types include:
    - Parallel Projection
    - Perspective Projection

Q46. What is JPEG?
Answer:-

- JPEG (Joint Photographic Experts Group) is a widely used image compression standard.
- It is mainly used for compressing photographic images.
- JPEG uses lossy compression, meaning some image quality is reduced to decrease file size.

Q47. Color choices in a 24-bit RGB system (512 × 512 frame buffer)
Answer:-

- Each pixel uses 24 bits (8 bits for Red, Green, and Blue each).
- Number of intensity levels per color = 28=256
- Total distinct colors =
  224=16,777,216 colors
- The resolution (512 × 512) affects the number of pixels, not the number of colors.

Q48. Difference between impact and non-impact printers

Answer:-

Basis	**Impact Printer**	**Non-Impact Printer**
Working	**Prints by striking paper**	**Prints without striking paper**
Noise	**Noisy**	**Quiet**
Speed	**Slower**	**Faster**
Quality	**Lower quality**	**Higher quality**
Examples	**Dot matrix printer**	**Inkjet, Laser printer**

Q49. What is a projector?

Answer:-

- A projector is an output device that displays images, videos, or data from a computer onto a large screen or surface.

- It is commonly used in presentations, classrooms, and home theaters.

Q50. What is hypermedia documents?

Answer:-

- Hypermedia documents are digital documents that include multiple media types such as:
    - Text
    - Images
    - Audio
    - Video
- They are interconnected using hyperlinks, allowing users to navigate non-linearly.

References

List of Books

Text Books

1. Computer Graphics

Authors: Hearn & Baker
Publisher: PHI

Brief: Comprehensive, detailed textbook covering Computer Graphics Discription

2. Computer Graphics-Principal & Practice

Authors: J.Foley, A.Vam Dam, S.Feiner, J.Hughes
Publisher: Pearson
Brief: Elementary book covering Principal and Practices of computer graphics .

Reference Books

1. Computer Graphics

Authors: Apurva A.Desai
Publisher: PHI
Brief: Focus on Graphics Primitives and problem-solving with proofs.

2. Computer Graphics

Author: Sinha and Udai

Publisher: Tata McGraw-Hill
Brief: Focous on Computer Graphics Realism.

Lab Quiz

Q1. Translation in 2D graphics is used to: CO3, L1

A. Change shape B. Change orientation

C. Change position D. Change size

Q2. The matrix representation of 2D translation requires: CO3, L2

A. 2 × 2 matrix B. 3 × 3 homogeneous matrix

C. 4 × 4 matrix D. 2 × 3 matrix

Q3. Homogeneous coordinates are used in computer graphics to: CO3, L2

A. Improve resolution B. Combine transformations

C. Increase memory D. Reduce computation

Q4. Composite transformations are achieved by: CO3, L2

A. Applying transformations independently B. Adding transformation matrices

C. Multiplying transformation matrices D. Reversing transformations

Q5. A point (2, 3) is translated by (4, -1). The new point is: CO3, L3

A. (6, 2) B. (−2, 4)

C. (4, 6) D. (3, 2)

Q6. Reflection about the Y-axis changes a point (x, y) to: CO3, L1

A. (x, −y) B. (−x, y)

C. (−x, −y) D. (y, x)

Q7. Shearing transformation primarily changes the: CO3, L1

A. Area B. Shape

C. Position D. Color

Q8. Window-to-viewport transformation is used to: CO3, L2

A. Clip objects B. Map world coordinates to screen coordinates

C. Rotate objects D. Fill polygons

Q9. Cohen–Sutherland algorithm is used for: CO3, L1

A. Polygon clipping B. Point clipping

C. Line clipping D. Curve clipping

Q10. Liang–Barsky line clipping algorithm is preferred because it: CO3, L2

A. Uses bit codes B. Uses parametric equations

C. Uses recursion D. Uses midpoint subdivision

Q11. A polygon surface is represented by: CO4, L1

A. Single point B. Line segments

C. Set of planar polygons D. Curved equations

Q12. Quadric surfaces include: CO4, L1

A. Sphere, cone, cylinder B. Triangle, rectangle

C. Bézier curves D. B-splines

Q13. Cubic spline interpolation ensures: CO4, L2

A. Discontinuous curves B. Smoothness at control points

C. Sharp corners D. Straight lines only

Q14. Bézier curves are controlled using: CO4, L1

A. Interpolation points B. Control points

C. Data points D. Boundary points

Q15. B-spline curves are preferred over Bézier curves because they: CO4, L2

A. Use fewer points B. Offer local control

C. Are simpler D. Are linear

Q16. In 3D graphics, transformation matrices are of size: CO4, L1

A. 2 × 2 B. 3 × 3

C. 4 × 4 D. 5 × 5

Q17. Which transformation preserves parallel lines? CO4, L3

A. Perspective projection B. Parallel projection

C. Rotation D. Scaling

Q18. Halftone patterns are used to: CO5, L1

A. Increase resolution B. Simulate shades of gray

C. Reduce computation D. Improve refresh rate

Q19. The RGB color model is based on: CO5, L1

A. Cyan, Magenta, Yellow B. Red, Green, Blue

C. Hue, Saturation, Value D. Luminance and Chrominance

Q20. The HSV color model is preferred for color selection because it: CO5, L2

A. Is device dependent B. Matches human perception

C. Uses primary colors D. Is hardware-oriented

****

Answer

Question No.	Correct Answer
Q1	C
Q2	B
Q3	B
Q4	C
Q5	A
Q6	B
Q7	B
Q8	B
Q9	C
Q10	B
Q11	C
Q12	A
Q13	B
Q14	B
Q15	B
Q16	C
Q17	B
Q18	B
Q19	B
Q20	B